MAB-Enhanced kNN for Image Anomaly Detection

We present an adaptive anomaly detection method that augments classical k-Nearest Neighbors (kNN) with a Multi-Armed Bandit (MAB) policy to select decision thresholds minimizing cost. On 2048-D embeddings, this approach improves recall and AUROC while reducing expected cost versus static thresholds.

Metric	Baseline	MAB-Enhanced
Recall	0.902	0.936
AUROC	0.852	0.878
min_cost	0.011	0.0095

Discussion: The MAB policy adaptively concentrates near thresholds that reduce expected cost while maintaining high recall. This bridges classical unsupervised scoring with reinforcement-inspired exploration.

Notebook: MAB_kNN_Anomaly_Detection.ipynb
GitHub Repo: github.com